

Faculty Profile

Name : Dr. Neelam Barak
Designation : Assistant Professor
Qualifications : B.Tech., M.Tech., Ph.D.
Phone : 9319963978
Email : neelam.barak@msit.in
Area of Interest/Specialization: Electronics& Communication, Robotics,
Optical microscopy, Digital Holography



Experience : 5+ years

Key Publications:

1. Enhanced tomographic imaging through scattering media via wavelength and exposure optimization in the SWIR digital holography, Optics and Lasers in Engineering, 2026
2. Simplified quantification of fluid flow using optimized laser speckle contrast imaging and statistical regression modelling, Measurement, 2025
3. Telecentric phase imaging at extended depth of focus using digital holographic microscopy, Precision Engineering, 2024
4. DMD Based Microscopic Fringe Projection Profilometry of Copper-Clad Substrates, MAPAN, 2024
5. Multimodal optical device to study dynamics of drying process, Optics and Lasers in Engineering, 2023
6. Module for zooming in extended depth of focus in digital holographic microscopy, Optics and Lasers in Engineering, 2023
7. Measurement of specular surfaces using electrically tunable lens in digital holography, Journal of Optics, 2022
8. Design and Development of an Automated Dual-Mode Microscopic System Using Electrically Tunable Lenses, Microscopy and Microanalysis, 2022
9. Simulation and analysis of variable numerical aperture wide-field microscopy, Micron, 2021

10. Statistical Modelling and Mapping of Intensity Spectrum in Breast MR Images, Mapan, 2021
11. Numerical four phase shifting microwave holography with phase step error analysis, IJRFMCAD, 2021
12. Automated extended depth of focus digital holographic microscopy using electrically tunable lens, Journal of Optics, 2020
13. Numerical three-step phase-shifting microwave holography, Optical Engineering, 2019
14. Two dimensional mapping by using single ultrasonic sensor, IJARCS, 2016

Papers presented in Conferences:

1. Motion detection, tracking and classification for automated Video Surveillance, ICPEICES, 2017
2. Two Wavelength Lensless Fourier Transform Digital Holographic Microscopy for Quantitative Phase Imaging with improved resolution, PHOTONICS 2018
3. Lens Less Fourier Transform Digital Holographic Microscopy for Quantitative Phase Imaging with Improved Resolution, Indo-US Colloquium on Recent Developments in Interdisciplinary Research, 2018.
4. Localization of sensor nodes using modified particle swarm optimization, ICACCI, 2016
5. Dual Wavelength Lensless Fourier Transform Digital Holographic Microscopy, IEEE INDICON, 2018

Awards and Recognitions:

1. Research Excellence Award awarded by Maharaja Surajmal Institute of Technology: 2024
2. GATE Qualified, Electronics & Communication Engg: 2014
3. CBSE Merit Certificate: 2010
4. Selected for QUTPRA Merit Scholarship by Queensland University of Technology, Australia: 2016

Patent/Copyright:

1. Telecentric Digital Holographic Microscopy System and Method for Extended Depth of Focus Imaging; Inventors: Dr. Vineeta Kumari, Ms. Neelam Barak, Dr. Gyanendra Sheoran, Prof. Dr. Ajay Kumar Sharma; Granted: 2024

2. Telecentric microscopic system for quantitative axial imaging and method thereof;
Inventors: Dr. Gyanendra Sheoran, Nishant Chhiller, Dr. Neelam Barak, Dr. Arun Anand,
Dr. Vineeta Kumari, Prof. Ajay K. Sharma, Published : 2026

Sponsored Project/Consultancy: N/A

Book Chapter/Books published:

1. Common Path Digital Holographic Microscopy Using Electrically Tunable Lens, Springer, 2021
2. Microscopy Using Liquid Lenses for Industrial and Biological Applications: Error Analysis and Uncertainty Evaluation, Handbook of Metrology and Applications, 2023
3. Necessity of Anatomically Real Numerical Phantoms in Optical Metrology: A Study, Handbook of Metrology and Applications, 2023

Ph.D Supervised: N/A

Memberships of Professional bodies:

1. Member, SPIE (2020)

Other Contributions: N/A